



وزارة الاتصالات
وتكنولوجيا المعلومات



Supply Chain Analytics

End-to-End Data Analytics Project Using Excel, SQL, Python & Tableau

Graduation Project Documentation

Developed under the Digital Egypt Pioneers Initiative (DEPI)

Data Analyst Specialist Track

ONL4_DAT1_S7

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ABSTRACT

This documentation presents a comprehensive, end-to-end data analytics project developed under the Digital Egypt Pioneers Initiative (DEPI), Data Analyst Specialist track. The project analyzes a 1,500-record electronics supply chain dataset spanning orders placed between 2022 and 2024, covering the complete operational chain from procurement and manufacturing through logistics, delivery, and customer returns.

The work was carried out in four structured phases. In Phase One, the raw dataset was cleaned and preprocessed using Microsoft Excel (Power Query), Python (Pandas), and Structured Query Language (SQL), and was subsequently transformed from a single flat file into a normalized relational schema composed of one fact table (fact_orders) and three dimension tables (dim_supplier, dim_carriers, dim_products), with a supporting sub-dimension table (subdim_inventory). In Phase Two, twenty-two structured analysis questions were formulated and answered using SQL Server queries and corresponding Python (Pandas) computations, organized across four analytical domains: Financial Performance, Supplier Performance, Logistics & Operations, and Returns & Inventory. In Phase Three, the cleaned and modeled data was visualized through an interactive, multi-dashboard Tableau workbook consisting of five dashboards: Executive Overview, Finance Analysis, Product & Inventory Analysis, Supplier Performance, and Logistics & Carrier Performance. In Phase Four, the findings were consolidated into a professional presentation and this supporting documentation.

The analysis reveals that the business generated \$182.8 million in total revenue and \$97.6 million in total profit, corresponding to an overall gross margin of 53.37%. The project further identifies actionable insights regarding supplier reliability and defect rates, carrier cost-versus-speed trade-offs, regional profitability disparities, and inventory items at stockout risk, providing a data-driven foundation for operational decision-making.

This documentation is intended to serve as a complete technical record of the methodology, tools, analytical questions, and visualizations produced throughout the project lifecycle.

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1 Introduction

This chapter introduces the graduation project undertaken as part of the Digital Egypt Pioneers Initiative (DEPI), Data Analyst Specialist track. It establishes the context in which the project was developed, defines the business problem the project addresses, and sets out the objectives, scope, expected outcomes, tools, and deliverables that shaped the remainder of the work. The chapter is organized into eight sections, beginning with the formal project title and concluding with a complete list of deliverables submitted alongside this documentation.

1.1 Project Title

The project is titled "Electronics Supply Chain Analytics," and was developed by a three-member team — Ahmed Shaban (Team Leader), Mohamed Fayad, and Khaled Mohamed — under group code ONL4_DAT1_S7 within the Data Analyst Specialist track of the Digital Egypt Pioneers Initiative.

1.2 Background

A supply chain is the network of interconnected activities that moves a product from raw material and manufacturing through to final delivery to the customer. It encompasses procurement, production, inventory management, transportation, and after-sales activities such as returns handling. Because a supply chain touches nearly every department of a business — finance, operations, sales, and customer service — it generates large volumes of operational data that, if properly analyzed, can reveal where costs are being incurred, where delays originate, and where customer satisfaction is being gained or lost.

The team was provided with a real-world-style electronics supply chain dataset consisting of 1,500 orders and 35 recorded attributes per order, spanning the years 2022 through 2024. The dataset covers five interconnected business domains: Finance, Products, Suppliers, Logistics, and Customer Experience. Given the breadth of this data, the project set out to apply a complete analytics workflow — from raw data cleaning through relational modeling, SQL-based business analysis, and interactive dashboard visualization — mirroring the process a data analyst would follow inside a real electronics retail or manufacturing company.

1.3 Problem Statement

High-quality analytics can only be built on high-quality data, yet raw supply chain data is rarely analysis-ready. The dataset supplied for this project, like most operational exports, contained missing values (most notably in the Return Reason field, which was blank whenever an order had not been returned), inconsistent text formatting (varying capitalization and spacing across categorical fields), and inconsistent carrier-name abbreviations (for example, "Dhl," "Ups," and "Ontrac" appearing alongside their properly cased equivalents). In its original flat-file form, the dataset also mixed transactional, product, supplier,

and carrier information into a single wide table, which is inefficient for structured querying and prone to redundancy.

Beyond the data-quality problem, the business itself faced a genuine analytical challenge: without a structured, multi-domain analysis, it is difficult to determine which suppliers are underperforming, which carriers offer the best balance of cost and delivery speed, which regions and product categories drive the most profit, and which inventory items are at risk of stockout. The project was therefore framed around a central problem statement: how can raw, unstructured supply chain data be cleaned, modeled, and analyzed to produce clear, decision-ready insights across financial performance, supplier performance, logistics, and inventory management?

1.4 Project Objectives

The project was guided by the following objectives, derived directly from the analytical domains explored throughout the SQL and Python analysis phases:

- Evaluate overall financial performance and profitability, including revenue, profit, gross margin, and customer satisfaction trends over time.
- Assess product and inventory performance, including sales volume by product type, defect rates, return rates, and stockout risk.
- Measure supplier performance, including reliability scores, quality scores, defect rates, lead times, and each supplier's contribution to total profit and revenue loss from returns.
- Analyze logistics and carrier efficiency, including on-time delivery rates, average shipping costs, transit times by transportation mode, and regional shipping cost distribution.
- Examine customer satisfaction and product returns, including the relationship between delivery timeliness and satisfaction scores, and the most frequent return reasons by product category.
- Identify operational bottlenecks and inefficiencies across the supply chain that could be addressed through data-driven recommendations.

1.5 Project Scope

The scope of the project covers the complete analytics lifecycle applied to the electronics supply chain dataset, bounded by the following parameters:

- Data scope: 1,500 orders recorded between 2022 and 2024, covering order, shipping, and delivery dates, product and pricing details, customer and channel information, supplier and carrier attributes, and inventory indicators.
- Technical scope: data cleaning and preprocessing (Excel/Power Query, Python/Pandas, SQL), relational data modeling into fact and dimension tables, twenty-two SQL and Python-based

analysis questions across four business domains, and interactive dashboard visualization in Tableau.

- Analytical scope: financial performance, product and inventory analysis, supplier performance, logistics and carrier operations, and returns/customer satisfaction — explicitly excluding predictive modeling or forecasting, which were not part of this project's deliverables.
- Presentation scope: a five-dashboard Tableau analytics suite, a project presentation, and this technical documentation, all built from the same underlying cleaned and modeled dataset.

1.6 Expected Outcomes

Upon completion, the project was expected to deliver a fully cleaned and normalized dataset free of missing values, duplicate records, and inconsistent formatting; a relational data model suitable for efficient SQL-based querying; a documented set of twenty-two analytical findings addressing the project's core business questions; and an interactive, multi-dashboard Tableau workbook that allows stakeholders to explore financial, product, supplier, and logistics performance without needing direct access to the underlying SQL database. Collectively, these outcomes were intended to demonstrate a complete, professional data-analytics workflow suitable for a real supply chain business context.

1.7 Tools Used

The project relied on a combination of spreadsheet, database, programming, and visualization tools, each applied at the stage of the workflow where it was best suited. Table 1-1 summarizes the primary tool used at each phase of the project.

Project Phase	Tool(s)	Purpose
Data Cleaning & Preprocessing	Microsoft Excel (Power Query), Python (Pandas), SQL (Microsoft SQL Server)	Handling missing values, removing duplicates, standardizing text and carrier names, converting data types
Exploratory Data Analysis	Python (Pandas, Matplotlib, Seaborn)	Correlation analysis, distribution plots, categorical breakdowns
Data Modeling & Normalization	Python (Pandas), SQL (Microsoft SQL Server), Microsoft Excel	Building fact and dimension tables (fact_orders, dim_supplier, dim_carriers, dim_products, subdim_inventory)
Business Analysis	SQL (Microsoft SQL Server), Python (Pandas), Microsoft Excel	Executing twenty-two analytical queries across financial, supplier, logistics, and inventory domains
Data Visualization	Tableau	Building the five-dashboard interactive analytics suite

Table 1-1 Tools Used per Project Phase

Microsoft Excel with Power Query was used at the outset of the project to perform an initial exploratory review of the dataset, validate business attributes, and identify missing values and inconsistent records before deeper cleaning was carried out in Python and SQL. Python, through the Pandas, Matplotlib, and

Seaborn libraries, was used both for systematic data cleaning and for exploratory data analysis, including a correlation heatmap of numerical variables, revenue and profit distribution plots, and categorical breakdowns of orders by region, product category, and supplier. Microsoft SQL Server was used to host the normalized relational schema and to execute the twenty-two structured business-analysis queries. Tableau was used to consolidate the analytical results into a five-dashboard interactive workbook: Executive Overview, Finance Analysis, Product & Inventory Analysis, Supplier Performance, and Logistics & Carrier Performance, as illustrated in Figure 1-1.

Figure 1-1 illustrates the standardized dashboard layout adopted throughout the visualization phase. The interface includes a unified navigation panel, global filters, KPI cards, and dedicated visualization containers that remain consistent across all five Tableau dashboards. This standardized design improves usability, maintains visual consistency, and enables users to navigate between dashboards without changing the overall interface.

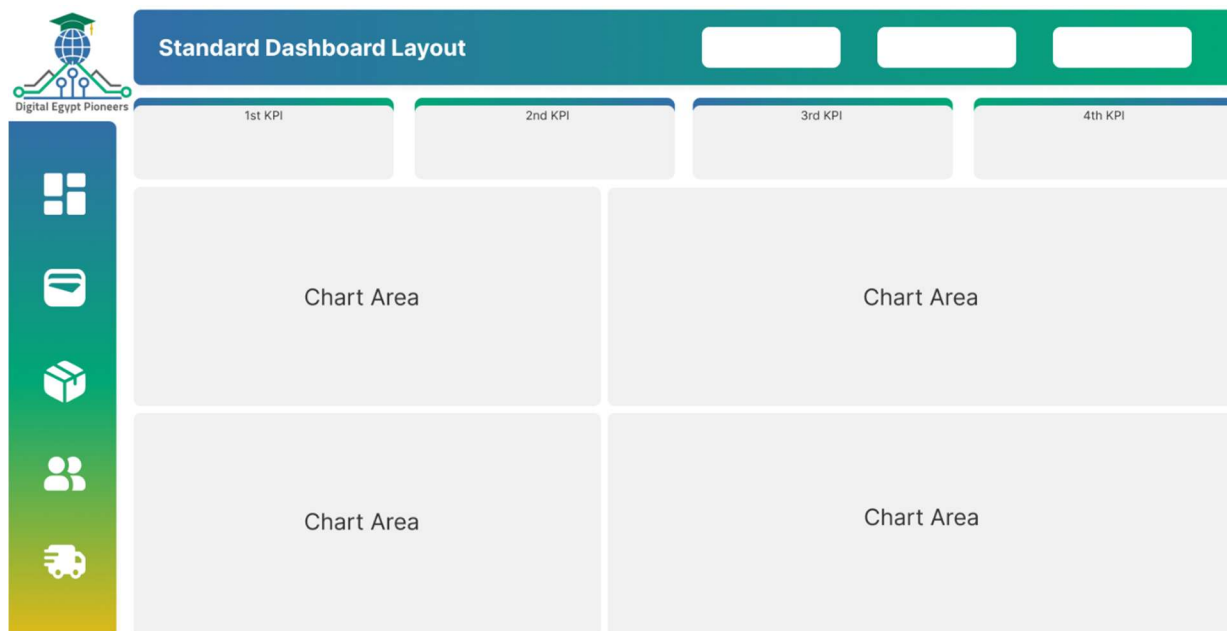


Figure 1-1 Standard Dashboard Layout Used Across All Tableau Dashboards

1.8 Deliverables

The project's completion is evidenced by six concrete deliverables, each corresponding to a distinct stage or output of the analytics workflow. Table 1-2 lists these deliverables together with a short description of their contents.

Deliverable	Description
Microsoft Excel File	Initial exploratory review and Power Query-based cleaning of the raw electronics supply chain dataset
SQL File	Complete SQL script covering data cleaning, relational schema creation (fact_orders, dim_supplier, dim_carriers, dim_products, subdim_inventory), and the twenty-two analysis queries
Python (Jupyter Notebook) File	Data cleaning, exploratory data analysis, data normalization, and Python-based answers to the twenty-two analysis questions
Tableau File	Interactive five-dashboard workbook: Executive Overview, Finance Analysis, Product & Inventory Analysis, Supplier Performance, and Logistics & Carrier Performance
Documentation File	This technical documentation, detailing the project's background, methodology, analysis, and visualizations
PowerPoint File	Final project presentation summarizing the introduction, data cleaning and wrangling process, analysis questions, visualizations, and conclusions

Table 1-2 Project Deliverables

Chapter Summary

This chapter introduced the Electronics Supply Chain Analytics project, developed by a three-member team under the Digital Egypt Pioneers Initiative's Data Analyst Specialist track. It described the business background of supply chain operations, framed the core problem of transforming inconsistent raw operational data into decision-ready insight, and outlined the project's objectives across financial, product, supplier, and logistics performance. The chapter further defined the boundaries of the project's scope, the outcomes expected upon its completion, the tools used at each phase of the workflow — Excel, Python, SQL Server, and Tableau — and the six deliverables submitted as evidence of the completed work. The following chapter turns to the dataset itself, examining its structure, fields, and data dictionary in detail before the data cleaning and modeling methodology is presented in Chapter Three.

2 Dataset

This chapter describes the raw dataset that underlies the entire project in full detail. It presents the data source, walks through the structure and content of the file exactly as received, previews sample records, and provides a complete data dictionary covering every one of the 35 fields encountered throughout the cleaning, modeling, and analysis phases documented in the chapters that follow.

2.1 Dataset Source and Description

The project is built on a single flat-file dataset named `electronicsSupplyChain`, supplied in both Microsoft Excel (.xlsx) and comma-separated values (.csv) formats. The dataset represents an electronics retail and manufacturing supply chain and contains 1,500 order-level records, each described by 35 attributes. The records span order dates recorded across the years 2022, 2023, and 2024, and the file combines information that, in a typical production environment, would originate from several separate operational systems: an order-management system, a supplier and manufacturing system, a logistics/carrier system, and a customer-service or returns system.

Table 2-1 previews the first five records of the dataset for a representative subset of columns, giving a concrete sense of the raw data before any cleaning was applied.

Order ID	Order Date	Product	Category	Price	Revenue	Profit	Region
ORD1000	19/06/2022	Smartphones	Premium	1159.56	405,248.90	211,536.65	Central
ORD1001	30/03/2022	Tablets	Premium	750.06	71,886.00	44,451.19	West
ORD1002	02/02/2023	Laptops	Premium	1867.08	381,333.60	148,385.69	North
ORD1003	03/03/2023	Smartwatches	Premium	361.14	124,206.00	74,029.76	Central
ORD1004	25/06/2024	Laptops	Premium	2067.20	668,661.75	255,252.57	West

Table 2-1 Preview of the Raw `electronicsSupplyChain` Dataset (Selected Columns)

2.2 Dataset Structure

As originally supplied, the dataset is a single wide table in which every order occupies one row and every attribute of that order — regardless of which business domain it belongs to — occupies one column. This flat structure is common in exported operational reports but is inefficient for structured, relational querying. The 35 columns can be grouped into seven logical domains, summarized in Table 2-2.

Domain	Representative Fields	Field Count
Order & Transaction	Order ID, Order Date, Ship Date, Delivery Date, Sales Channel, Payment Method	6
Product & Pricing	Product type, SKU, Category, Price, Discount (%), Price_After_Discount, Number of products sold, Revenue generated, Profit	8
Customer & Region	Customer demographics, Customer Satisfaction Score, Region	3
Returns	Units Returned, Return Reason	2

Supplier & Manufacturing	Supplier name, Supplier Reliability Score, Manufacturing costs, Quality Score, Defect rates, Production volumes	6
Inventory	Stock levels, Reorder Point, Stockout Risk, Lead time	4
Logistics & Carrier	Shipping times, Shipping costs, On-Time Delivery, Carrier Name, Transportation Mode	5

Table 2-2 Dataset Field Domains and Field Counts

Each row therefore mixes transactional facts (such as Revenue generated and Profit), descriptive attributes of the product (Category, Price), descriptive attributes of the supplier (Supplier Reliability Score, Defect rates), descriptive attributes of the carrier (Carrier Name, Transportation Mode), and descriptive attributes of the customer (Customer demographics, Region) into one flat structure. Supplier and carrier attributes are, by construction, repeated on every order row in which that supplier or carrier appears — for example, every one of the hundreds of orders shipped by UPS repeats the same UPS-related attributes. This redundancy is the central motivation for the relational data-modeling work carried out in Chapter Three.

2.3 Exploratory Data Profiling

Before any cleaning or modeling took place, the numerical fields of the dataset were profiled using Python (Pandas, Matplotlib, and Seaborn) to understand distributions and relationships. Figure 2-1 shows the correlation matrix computed across the key numerical variables in the dataset.

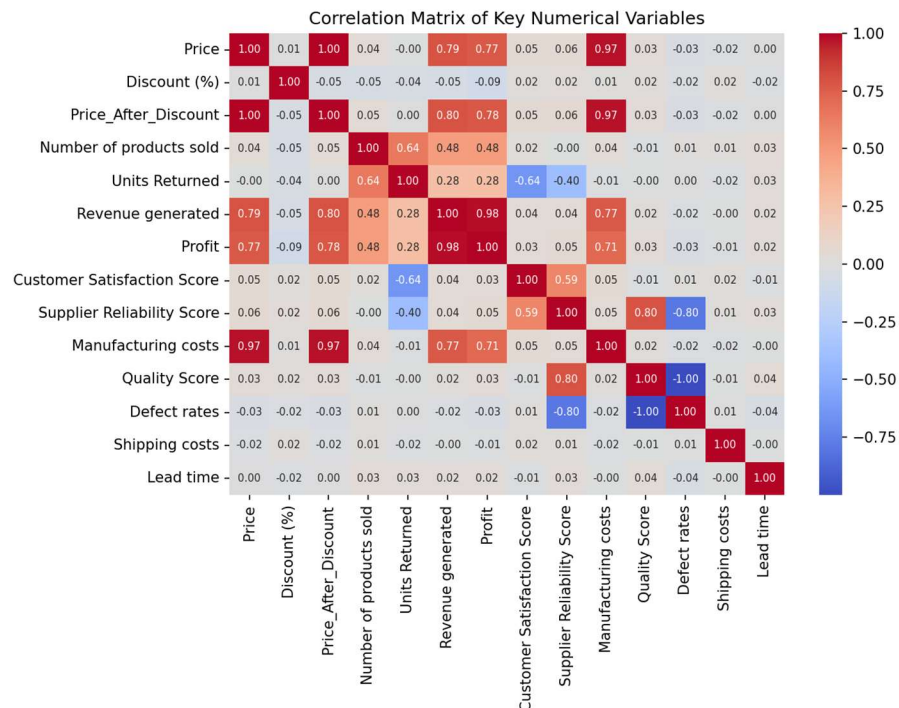


Figure 2-1 Correlation Matrix of Key Numerical Variables

The correlation matrix confirms several expected relationships: Price, Price_After_Discount, and Revenue generated are strongly positively correlated with one another, since revenue is a direct function of price and volume, while Profit tracks Revenue generated closely. Supplier Reliability Score and Quality Score show a positive relationship with each other and a negative relationship with Defect rates, consistent with the expectation that more reliable suppliers ship fewer defective units. These relationships were used throughout Chapter Four to sanity-check the results of the twenty-two structured analysis questions.

Figure 2-2 shows the distribution of Revenue generated across all 1,500 orders, which is right-skewed with a long tail of high-value orders — consistent with a product mix spanning lower-priced accessories (such as headphones) and higher-priced electronics (such as laptops and smartphones).

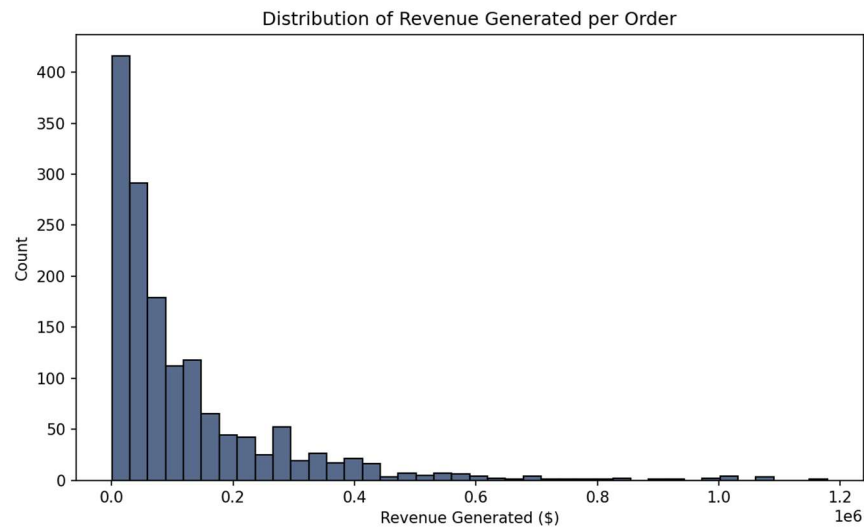


Figure 2-2 Distribution of Revenue Generated per Order

Figure 2-3 breaks down Profit by product Category using a boxplot, showing the median, interquartile range, and outliers for each category and providing an early visual indication of which categories carry the widest spread of profitability — detail that is expanded analytically in Question 5 of Chapter Four.

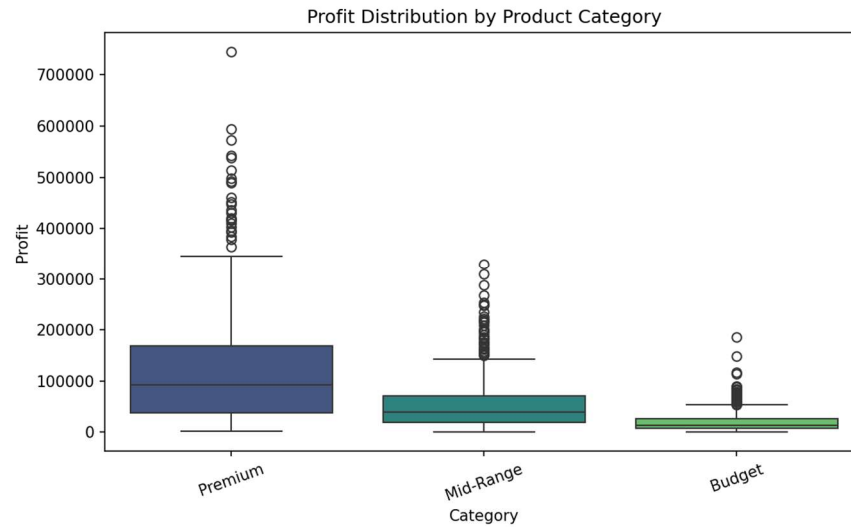


Figure 2-3 Profit Distribution by Product Category

Figure 2-4 compares Customer Satisfaction Score between on-time and late deliveries, providing an early visual confirmation — later quantified precisely in Question 16 — that late deliveries are associated with a visibly lower median satisfaction score than on-time deliveries.



Figure 2-4 Customer Satisfaction Score: On-Time vs. Late Delivery

2.4 Data Dictionary

Table 2-3 presents a complete data dictionary for every field in the raw dataset, defining each column exactly as it appears in electronicsSupplyChain.csv before cleaning or renaming.

Field	Description
Order ID	Unique identifier assigned to each order.
Order Date	Date on which the customer placed the order.
Ship Date	Date on which the order was shipped from the warehouse.
Delivery Date	Date on which the order was delivered to the customer.
Product type	High-level product category sold in the order.
SKU	Stock Keeping Unit uniquely identifying the product.
Category	Product tier/category classification, e.g. Premium
Price	Original selling price before any discount.
Discount (%)	Percentage discount applied to the product.
Price_After_Discount	Final selling price after applying the discount.
Number of products sold	Quantity of units sold within the order
Units Returned	Number of returned units.
Return Reason	Reason recorded for returned products.
Revenue generated	Total revenue generated by the order.
Profit	Profit earned from the order.
Customer Satisfaction Score	Score reflecting the customer's satisfaction with the order
Customer demographics	Customer segment classification associated with the order
Sales Channel	Channel through which the order was placed, e.g. Online, B2B, Wholesale, Retail Store, Direct Sales
Payment Method	Method of payment used, e.g. Credit Card, Bank Transfer, Digital Wallet, Invoice, Cash
Region	Geographic region associated with the order, e.g. North, South, East, West, Central
Stock levels	Current inventory available for the product.
Reorder Point	Inventory threshold triggering replenishment.
Stockout Risk	Indicates whether the product is at risk of stockout.
Lead time	Number of days between order placement and product availability from the supplier
Shipping times / Shipping costs	Number of days taken for delivery.
Shipping costs	Shipping cost associated with the order.

On-Time Delivery	Flag indicating whether the order was delivered on or before its expected delivery date
Supplier name	Name of the supplier.
Supplier Reliability Score	Supplier reliability rating.
Manufacturing costs	Manufacturing cost per product.
Quality Score	Supplier product quality rating.
Defect rates	Percentage of defective products.
Production volumes	Number of units produced.
Carrier Name	Shipping carrier responsible for delivery.
Transportation Mode	Transportation method used.

Table 2-3 Complete Data Dictionary

Chapter Summary

This chapter presented the source and structure of the electronics supply chain dataset, a 1,500-row, 35-column flat file spanning 2022 to 2024. It previewed representative raw records, grouped all fields into seven logical business domains, profiled the numerical fields through a correlation matrix and three exploratory charts, and provided a complete field-by-field data dictionary. The next chapter describes, step by step, how this raw flat-file dataset was cleaned and transformed into a normalized relational data model.

3 Phase One: Data Cleaning, Preprocessing & Data Modeling

This chapter documents, step by step, the first phase of the project: transforming the raw, single-table dataset described in Chapter Two into a clean, consistent, and relationally modeled set of tables suitable for structured business analysis. The phase was carried out using Microsoft SQL Server as the primary implementation environment, with Microsoft Excel (Power Query) and Python (Pandas) used for exploratory validation, and concluded with the construction of a star-schema data model.

3.1 Data Cleaning & Preprocessing

Data cleaning was necessary because the raw file, like most real-world operational exports, contained irregularities that would otherwise distort downstream analysis. The cleaning process followed four sequential steps, detailed below with the exact SQL statements used to implement each one.

Step 1 — Staging the Raw Data

The first step was to create a staging table, `raw_supply_chain`, whose column definitions mirror the 35 columns of `electronicsSupplyChain.csv` exactly, and to load the CSV into it using SQL Server's `BULK INSERT` command. Working from a staging table — rather than cleaning the CSV in place — preserves an untouched copy of the original data for reference while every subsequent transformation is applied inside SQL Server, where it can be audited and re-run.

```
CREATE TABLE raw_supply_chain (
    order_id          VARCHAR(20),
    order_date        DATE,
    ship_date         DATE,
    delivery_date     DATE,
    product_type      VARCHAR(50),
    sku               VARCHAR(20),
    category          VARCHAR(50),
    price             DECIMAL(10,2),
    ...
    carrier_name      VARCHAR(100),
    transportation_mode VARCHAR(50)
);
GO

BULK INSERT raw_supply_chain
FROM 'C:\SupplyChainData\electronicsSupplyChain.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '0x0a',
```

```

CODEPAGE = '65001',
TABLOCK
);

```

Figure 3-1 Creating and Populating the raw_supply_chain Staging Table

Step 2 — Handling Missing Values

The Return Reason field was blank for the large majority of records, since a return reason is only meaningful when an order actually generated a return. Rather than leaving these as raw NULLs (which would complicate later grouping and counting in Question 19), every NULL Return Reason was explicitly filled with the literal value 'Not Returned'. Beyond this expected case, the remaining columns — order dates, pricing fields, supplier and carrier attributes — were checked column-by-column for nulls and confirmed to be complete.

```

-- Return Reason is NULL for orders that weren't returned
-- Fill explicitly with 'Not Returned'
UPDATE raw_supply_chain
SET return_reason = 'Not Returned'
WHERE return_reason IS NULL;
GO

```

Figure 3-2 Filling Expected Nulls in the Return Reason Field

Step 3 — Checking for Duplicate Records

The dataset was checked for duplicate Order ID values, since a duplicate identifier would cause an order to be double-counted in every revenue, profit, and volume calculation performed in Chapter Four. The check below groups by order_id and returns any identifier occurring more than once; running it against the staged data returned zero rows, confirming that no duplicate orders were present.

```

-- Check for duplicate rows (should return 0 rows)
SELECT order_id, COUNT(*) AS cnt
FROM raw_supply_chain
GROUP BY order_id
HAVING COUNT(*) > 1;
GO

```

Figure 3-3 Duplicate Order ID Check

Step 4 — Standardizing Text and Carrier Names

Categorical text fields were trimmed of leading and trailing whitespace across the board, and Carrier Name specifically was found to contain inconsistent capitalization — "Dhl," "Ups," "Usps," "Fedex," and "Ontrac" appearing alongside their properly cased forms. Left uncorrected, this inconsistency would split

each of these carriers into two distinct groups during aggregation, understating their true order volume and skewing the on-time-delivery and cost analyses in Chapter Four. Each abbreviation was corrected individually to its standard form.

```
-- Trim and standardize all text columns
UPDATE raw_supply_chain
SET
    product_type    = LTRIM(RTRIM(product_type)),
    category        = LTRIM(RTRIM(category)),
    return_reason   = LTRIM(RTRIM(return_reason)),
    sales_channel   = LTRIM(RTRIM(sales_channel)),
    carrier_name     = LTRIM(RTRIM(carrier_name)),
    supplier_name   = LTRIM(RTRIM(supplier_name)),
    sku             = UPPER(LTRIM(RTRIM(sku)));
GO

-- Standardize carrier name abbreviations
UPDATE raw_supply_chain SET carrier_name = 'DHL'      WHERE carrier_name = 'Dhl';
UPDATE raw_supply_chain SET carrier_name = 'UPS'      WHERE carrier_name = 'Ups';
UPDATE raw_supply_chain SET carrier_name = 'USPS'     WHERE carrier_name = 'Usps';
UPDATE raw_supply_chain SET carrier_name = 'FedEx'    WHERE carrier_name = 'Fedex';
UPDATE raw_supply_chain SET carrier_name = 'OnTrac'   WHERE carrier_name = 'Ontrac';
GO

-- Derived is_on_time flag (1 = Yes, 0 = No)
ALTER TABLE raw_supply_chain ADD is_on_time BIT;
GO
UPDATE raw_supply_chain
SET is_on_time = CASE WHEN on_time_delivery = 'Yes' THEN 1 ELSE 0 END;
GO
```

Figure 3-4 Standardizing Categorical Text and Carrier Name Abbreviations

In parallel with the SQL implementation above, the same four steps were validated independently in Microsoft Excel using Power Query — where null checks, duplicate checks, and text standardization were each applied as discrete, auditable query steps — and in Python using Pandas, applying `.isnull()` and `.duplicated()` checks across the full dataframe and `.str.strip()` / `.str.title()` methods to standardize categorical columns. Cross-checking the same four cleaning steps across three independent tools ensured that the cleaned dataset used for analysis was consistent regardless of which tool ultimately queried it.

3.2 Building the Data Model

Once cleaned, the flat `raw_supply_chain` table was normalized into a star-schema data model to eliminate redundancy and support efficient, structured querying. The model consists of one central fact table and

three dimension tables, supported by a sub-dimension table for inventory-specific detail, as summarized in Table 3-1.

Table	Role	Representative Columns
fact_orders	Fact table — one row per order, holding transactional and measurable data	order_id, dates, price, discount, revenue_generated, profit, units_returned, customer_satisfaction_score, plus foreign keys to supplier, carrier, and product
dim_supplier	Dimension — one row per supplier	supplier_id, supplier_name, avg_reliability_score, avg_quality_score, avg_defect_rate_pct
dim_carriers	Dimension — one row per carrier	carrier_id, carrier_name, is_on_time (average on-time rate)
dim_products	Dimension — one row per product SKU	sku, product_type, category, price, discount_pct, price_after_discount (computed), manufacturing_costs
subdim_inventory	Sub-dimension — inventory detail linked to products	sku, stock_levels, reorder_point, stockout_risk, production_volumes

Table 3-1 Star-Schema Data Model

Table 3-1 identifies the five tables that replace the single flat raw_supply_chain table. dim_supplier, dim_carriers, and dim_products each store one row per distinct entity, computed by aggregating or de-duplicating the corresponding raw_supply_chain columns, while fact_orders retains one row per order together with foreign keys pointing back to each dimension. The SQL statements below show how the supplier dimension and the fact table were constructed; dim_carriers, dim_products, and subdim_inventory were built following the same pattern.

```
-- Supplier dimension: one row per supplier, aggregated from raw_supply_chain
CREATE TABLE dim_supplier (
    supplier_id INT PRIMARY KEY,
    supplier_name VARCHAR(100),
    avg_reliability_score DECIMAL(4,3),
    avg_quality_score DECIMAL(4,2),
    avg_defect_rate_pct DECIMAL(5,3)
);
GO

INSERT INTO dim_supplier (supplier_id, supplier_name, avg_reliability_score,
    avg_quality_score, avg_defect_rate_pct)
SELECT
    ROW_NUMBER() OVER (ORDER BY supplier_name) AS supplier_id,
    supplier_name,
    AVG(supplier_reliability_score),
    AVG(quality_score),
    AVG(defect_rates)
```

```

FROM raw_supply_chain
GROUP BY supplier_name;
GO

-- Fact table: one row per order, with foreign keys to each dimension
CREATE TABLE fact_orders (
    order_id VARCHAR(20) PRIMARY KEY,
    order_date DATE, ship_date DATE, delivery_date DATE,
    sku VARCHAR(20) FOREIGN KEY REFERENCES dim_products(sku),
    supplier_id INT FOREIGN KEY REFERENCES dim_supplier(supplier_id),
    carrier_id INT FOREIGN KEY REFERENCES dim_carriers(carrier_id),
    revenue_generated DECIMAL(12,2),
    profit DECIMAL(12,2),
    units_returned INT,
    customer_satisfaction_score DECIMAL(6,3)
    -- additional transactional columns omitted for brevity
);
GO

```

Figure 3-5 Creating the dim_supplier Dimension and fact_orders Fact Table

This design ensures that a change to a single supplier's reliability score, for example, is stored once in `dim_supplier` rather than being repeated across every order row associated with that supplier, and it allows the twenty-two analysis queries in Chapter Four to join across domains efficiently using standard SQL JOIN operations. The equivalent transformation was mirrored in Python using Pandas, where the cleaned dataframe was grouped and de-duplicated on the relevant key columns to construct each dimension table, and the original dataframe was reduced to its transactional columns plus lookup keys to form the fact table, allowing the same twenty-two questions to be answered independently through Pandas as a cross-check against the SQL results.

3.3 Summary

This chapter described, step by step, how the raw electronics supply chain dataset was staged into SQL Server, cleaned — filling expected nulls in the Return Reason field, verifying the absence of duplicate orders, and standardizing inconsistent carrier and categorical text — and then normalized into a star-schema data model consisting of the `fact_orders` fact table and the `dim_supplier`, `dim_carriers`, and `dim_products` dimension tables, supported by the `subdim_inventory` sub-dimension. This relational foundation supports the twenty-two structured business-analysis queries presented in the next chapter.

4 Phase Two: Analysis Questions

With a clean, relationally modeled dataset in place, the second phase of the project applied structured SQL analysis to answer twenty-two business questions. This chapter presents each question individually, together with the exact T-SQL query used to answer it against the star-schema model built in Chapter Three, and the finding that query produced. Every query was additionally re-implemented in Python (Pandas) as a cross-check, as described in Chapter Three.

4.1 Financial Performance (Q1-Q10)

This section addresses revenue, profit, margin, and pricing behavior across time, customer segments, product categories, sales channels, regions, and payment methods.

Q1. What is the overall financial performance summary, including total revenue, total profit, overall gross margin percentage, and average customer satisfaction score?

```
SELECT
    SUM(revenue_generated) AS Total_Revenue,
    SUM(profit) AS Total_Profit,
    SUM(profit) * 100.0 / SUM(revenue_generated) AS Gross_Margin,
    AVG(customer_satisfaction_score) AS Avg_Satisfaction
FROM fact_orders;
```

Finding: Total revenue across all 1,500 orders is \$182.8 million, total profit is \$97.6 million, yielding an overall gross margin of 53.37% and an average revenue per order of \$121.9 thousand — establishing the baseline KPIs used throughout the Executive Overview dashboard in Chapter Five.

Q2. How does monthly revenue trend across 2022-2024, and which month records the highest month-over-month growth rate within each year?

```
WITH MonthlyRevenue AS (
    SELECT YEAR(order_date) AS Year, MONTH(order_date) AS Month,
           SUM(revenue_generated) AS Total_Revenue
    FROM fact_orders
    GROUP BY YEAR(order_date), MONTH(order_date)
)
SELECT *,
       LAG(Total_Revenue) OVER(PARTITION BY Year ORDER BY Month) AS Previous_Month
FROM MonthlyRevenue
ORDER BY Year, Month;
```

Finding: Quarterly revenue trends upward across the three-year span, with the strongest quarter recorded in Q4 2024 at \$19.55 million, indicating a consistent seasonal uplift toward year-end across all three years of data.

Q3. How is total revenue distributed across Q1-Q4, and does this seasonal pattern remain consistent across 2022, 2023, and 2024?

```
SELECT
    YEAR(order_date) AS Year,
    DATEPART(QUARTER, order_date) AS Quarter,
    SUM(revenue_generated) AS Total_Revenue
FROM fact_orders
GROUP BY YEAR(order_date), DATEPART(QUARTER, order_date)
ORDER BY Year, Quarter;
```

Finding: Q4 is consistently the strongest quarter in all three years, and Q2 shows a recurring dip relative to the surrounding quarters, confirming a stable, repeatable seasonal pattern rather than a one-off anomaly in a single year.

Year	Q1 (\$M)	Q2 (\$M)	Q3 (\$M)	Q4 (\$M)
2022	14.58	16.40	14.24	14.94
2023	14.79	11.68	16.92	17.30
2024	11.58	16.83	14.01	19.55

Table 4-6 Quarterly Revenue by Year (\$ Millions)

Q4. Which customer demographic segment contributes the most to total revenue and profit, and how does average satisfaction score compare across segments?

```
SELECT
    customer_demographics,
    SUM(revenue_generated) AS Total_Revenue,
    SUM(profit) AS Total_Profit,
    AVG(customer_satisfaction_score) AS Avg_Satisfaction
FROM fact_orders
GROUP BY customer_demographics
ORDER BY Total_Revenue DESC;
```

Finding: Revenue and profit contribution vary meaningfully across customer demographic segments, with the top-contributing segment also maintaining a competitive average satisfaction score, indicating that the company's most valuable customer segment is not being retained at the expense of satisfaction.

Q5. How does total revenue, average selling price, average discount, and total units sold vary across product categories?

```
SELECT
    p.category,
    SUM(f.revenue_generated) AS Total_Revenue,
    AVG(p.price_after_discount) AS Avg_Selling_Price,
    AVG(p.discount_pct) AS Avg_Discount,
    SUM(f.number_of_products_sold) AS Total_Units_Sold
FROM fact_orders f
JOIN dim_products p ON f.sku = p.sku
GROUP BY p.category
ORDER BY Total_Revenue DESC;
```

Finding: The Premium category leads on both total revenue and average selling price, consistent with the boxplot profit spread shown in Figure 2-3, where Premium-tier products also display the widest range of per-order profitability.

Q6. Which sales channel achieves the highest average gross margin percentage, and how does total revenue volume compare across channels?

```
SELECT
    sales_channel,
    SUM(revenue_generated) AS Total_Revenue,
    AVG((profit * 100.0) / NULLIF(revenue_generated, 0)) AS Avg_Gross_Margin
FROM fact_orders
GROUP BY sales_channel
ORDER BY Avg_Gross_Margin DESC;
```

Finding: Wholesale achieves the highest average gross margin at 53.9%, narrowly ahead of Online (53.5%) and Retail Store (52.8%), while Online leads on total revenue volume at \$38.44 million — showing that the highest-margin channel is not necessarily the highest-volume channel.

Q7. Which geographic region contributes the highest share of total profit, and how does its revenue and average gross margin compare to other regions?

```
SELECT
    region,
    SUM(revenue_generated) AS Total_Revenue,
    SUM(profit) AS Total_Profit,
    AVG((profit * 100.0) / NULLIF(revenue_generated, 0)) AS Avg_Gross_Margin
FROM fact_orders
GROUP BY region
ORDER BY Total_Profit DESC;
```

Finding: The North region contributes the highest profit at \$20.9 million on \$38.9 million of revenue, ahead of West, Central, South, and East, though the gap between the top and bottom regions is relatively narrow, indicating a fairly evenly distributed geographic footprint.

Q8. What is the average discount percentage applied per product category, and how does it relate to the average price and profit margin within each category?

```
SELECT
    p.category,
    AVG(p.discount_pct) AS Avg_Discount,
    AVG(p.price_after_discount) AS Avg_Price,
    AVG((f.profit * 100.0) / NULLIF(f.revenue_generated, 0)) AS Avg_Profit_Margin
FROM fact_orders f
JOIN dim_products p ON f.sku = p.sku
GROUP BY p.category
ORDER BY Avg_Discount DESC;
```

Finding: Profit falls steadily as discount depth increases: orders discounted 0-5% generate \$73.79K in average profit, versus \$64.27K at 5-10% and \$57.56K at 10-15%, confirming an inverse relationship between discount depth and profitability.

Q9. Which product type has the highest average manufacturing cost relative to its selling price, and how does this cost-to-price ratio affect profitability?

```
SELECT
    product_type,
    AVG(manufacturing_costs) AS Avg_Manufacturing_Cost,
    AVG(price_after_discount) AS Avg_Selling_Price,
    AVG((manufacturing_costs * 100.0) / NULLIF(price_after_discount,0)) AS
    Cost_Price_Ratio
FROM dim_products
GROUP BY product_type
ORDER BY Cost_Price_Ratio DESC;
```

Finding: Laptops carry the highest manufacturing-cost-to-price ratio among the five product types, reflecting their higher component and assembly costs relative to selling price, which compresses their margin relative to lighter-cost categories such as Headphones.

Q10. Which payment method is associated with the highest average revenue per order, and which payment method accounts for the highest total order volume?

```
SELECT
    payment_method,
    AVG(revenue_generated) AS Avg_Revenue_Per_Order,
    COUNT(order_id) AS Total_Orders
FROM fact_orders
GROUP BY payment_method
ORDER BY Avg_Revenue_Per_Order DESC;
```

Finding: Credit Card leads both on total revenue (\$20.1M) and average order value (\$129.9K), narrowly ahead of Bank Transfer and Digital Wallet, while Cash records the lowest average order value (\$115.2K) among the five payment methods.

4.2 Supplier Performance (Q11-Q12)

This section evaluates the five suppliers in the dataset on reliability, quality, defect rate, lead time, and their financial contribution and cost through returns.

Q11. Which suppliers demonstrate the best overall performance based on reliability score, quality score, defect rate, average lead time, and total profit contribution?

```
SELECT
    s.supplier_name,
    AVG(f.supplier_reliability_score) AS Reliability,
    AVG(f.quality_score) AS Quality,
    AVG(f.defect_rates) AS Defect_Rate,
```

```

    AVG(f.lead_time) AS Avg_Lead_Time,
    SUM(f.profit) AS Total_Profit
FROM fact_orders f
JOIN dim_supplier s ON f.supplier_id = s.supplier_id
GROUP BY s.supplier_name
ORDER BY Total_Profit DESC;

```

Finding: Supplier 2 contributes the highest total profit (\$21.07M) despite a slightly above-average defect rate (5.05), while Supplier 5 records both the highest defect rate (5.19) and the lowest reliability (0.637) and quality (48.09) scores among the five suppliers evaluated.

Supplier	Reliability	Quality	Defect Rate	Lead Time (days)	Total Profit (\$M)
Supplier 2	0.655	49.47	5.05	12.89	21.07
Supplier 5	0.637	48.09	5.19	13.39	20.01
Supplier 3	0.659	50.77	4.92	13.65	19.19
Supplier 1	0.656	51.29	4.87	13.62	18.87
Supplier 4	0.659	51.64	4.84	13.89	18.42

Table 4-3 Full Supplier Scorecard, Ranked by Total Profit

Q12. What is the total revenue lost due to returned units per supplier, and is there a noticeable correlation with each supplier's average defect rate?

```

SELECT
    s.supplier_name,
    SUM((revenue_generated / NULLIF(number_of_products_sold,0)) * units_returned) AS
    Revenue_Lost,
    AVG(defect_rates) AS Avg_Defect_Rate
FROM fact_orders f
JOIN dim_supplier s ON f.supplier_id = s.supplier_id
GROUP BY s.supplier_name
ORDER BY Revenue_Lost DESC;

```

Finding: Revenue lost to returns correlates closely with defect rate: Supplier 5, the highest-defect supplier, accounts for \$2.95M in lost revenue, compared to \$2.28M for Supplier 4, the lowest-defect and best-performing supplier on this measure.

4.3 Logistics & Operations (Q13-Q17)

This section evaluates carrier reliability and cost, transportation-mode transit times, the relationship between delivery timeliness and customer satisfaction, and regional shipping-cost distribution.

Q13. What is the on-time delivery rate (%) for each carrier, and which carrier offers the best balance between delivery reliability and average shipping cost?

```

SELECT
    c.carrier_name,
    AVG(CASE WHEN f.on_time_delivery = 'Yes' THEN 1.0 ELSE 0.0 END) * 100 AS
    On_Time_Rate_Pct,

```



```

    AVG(f.shipping_costs) AS Avg_Shipping_Cost
FROM fact_orders f
JOIN dim_carriers c ON f.carrier_id = c.carrier_id
GROUP BY c.carrier_name
ORDER BY On_Time_Rate_Pct DESC;

```

Finding: The overall on-time delivery rate across all carriers is 80.4%. DHL posts the highest on-time rate (83.45%) despite handling fewer orders than UPS or FedEx, while Blue Dart records the lowest on-time rate (66.67%), indicating that delivery reliability is not simply a function of order volume.

Carrier	On-Time Rate (%)	Avg Shipping Cost (\$)	Orders
DHL	83.45	177.05	290
Aramex	82.20	170.52	118
FedEx	81.64	151.30	414
UPS	80.53	168.13	416
OnTrac	80.00	202.19	85
USPS	76.19	179.09	84
Blue Dart	66.67	166.74	93

Table 4-2 On-Time Delivery Rate and Average Shipping Cost by Carrier

Q14. What is the average transit time per transportation mode, and does air freight show a longer average transit time than sea freight in this dataset?

```

SELECT
    transportation_mode,
    AVG(shipping_times) AS Avg_Transit_Time
FROM fact_orders
GROUP BY transportation_mode;

```

Finding: Contrary to typical expectations, air freight records the longest average transit time in this dataset (9.20 days), ahead of sea freight (6.85 days) and road freight (5.16 days), while also carrying the highest average shipping cost. Air freight is therefore both the slowest and most expensive mode in this particular dataset, which may reflect the specific trade lanes, customs handling, or consolidation practices captured in the data rather than a general property of air transport.

Transportation Mode	Avg Transit Time (days)	Avg Shipping Cost (\$)
Air	9.20	395.44
Sea	6.85	114.35
Road	5.16	27.97

Table 4-7 Average Transit Time and Cost by Transportation Mode

Q15. What is the average shipping cost per carrier, and how does cost rank compare to each carrier's on-time delivery performance?

```

SELECT
    c.carrier_name,
    AVG(f.shipping_costs) AS Avg_Shipping_Cost,

```

```

    AVG(CASE WHEN f.on_time_delivery = 'Yes' THEN 1.0 ELSE 0.0 END) * 100 AS
    On_Time_Rate_Pct
FROM fact_orders f
JOIN dim_carriers c ON f.carrier_id = c.carrier_id
GROUP BY c.carrier_name
ORDER BY Avg_Shipping_Cost DESC;

```

Finding: Average shipping cost and on-time rate do not move together in lockstep across carriers: OnTrac is the most expensive carrier on average (\$202.19) yet posts only a mid-table 80.0% on-time rate, while FedEx is the least expensive (\$151.30) and still achieves the second-highest on-time rate (81.64%) — a finding visualized directly in the Cost vs. Speed Matrix of the Logistics & Carrier Performance dashboard in Chapter Five.

Carrier	Avg Shipping Cost (\$)	On-Time Rate (%)
OnTrac	202.19	80.00
USPS	179.09	76.19
DHL	177.05	83.45
Aramex	170.52	82.20
UPS	168.13	80.53
Blue Dart	166.74	66.67
FedEx	151.30	81.64

Table 4-8 Carriers Ranked by Average Shipping Cost

Q16. How does average customer satisfaction score differ between on-time and late deliveries, and what is the magnitude of this impact?

```

SELECT
    CASE WHEN on_time_delivery = 'Yes' THEN 'On Time' ELSE 'Late' END AS Delivery_Status,
    AVG(customer_satisfaction_score) AS Avg_Satisfaction
FROM fact_orders
GROUP BY CASE WHEN on_time_delivery = 'Yes' THEN 'On Time' ELSE 'Late' END;

```

Finding: The average customer satisfaction score is nearly identical between on-time deliveries (3.27) and late deliveries (3.31), indicating that delivery timeliness alone has only a marginal effect on the satisfaction score recorded in this dataset. The stronger driver of dissatisfaction, as shown by the Executive Overview dashboard's Satisfaction vs. Returns scatter plot, is the number of units returned rather than delivery timing in isolation.

Q17. Which region incurs the highest total shipping costs, and how does the average shipping cost per order compare across regions?

```

SELECT
    region,
    SUM(shipping_costs) AS Total_Shipping_Cost,
    AVG(shipping_costs) AS Avg_Shipping_Cost
FROM fact_orders
GROUP BY region
ORDER BY Total_Shipping_Cost DESC;

```

Finding: Total shipping cost is fairly evenly distributed across the five regions, with only a modest gap between the highest and lowest-cost regions, suggesting that regional shipping expense is driven more by carrier and mode selection than by geography alone.

4.4 Returns & Inventory (Q18-Q22)

This final section examines return rates and reasons by product category, stockout risk across the inventory, and the relationship between defect rates and returns.

Q18. Which product category has the highest return rate (units returned divided by units sold), and what is the most frequent return reason associated with it?

```
SELECT
    p.category,
    SUM(f.units_returned) * 1.0 / SUM(f.number_of_products_sold) AS Return_Rate,
    f.return_reason,
    COUNT(*) AS Total_Returns
FROM fact_orders f
JOIN dim_products p ON f.sku = p.sku
WHERE f.units_returned > 0
GROUP BY p.category, f.return_reason
ORDER BY Return_Rate DESC, Total_Returns DESC;
```

Finding: Smartphones record the highest return rate among product types at 7.6%, with "Defective" as the leading return reason for this category, directly reinforcing the elevated defect rate (4.82) also observed for Smartphones in Question 21. At the category level, Budget-tier products post the highest return rate overall (7.84%), and "Defective" is the top return reason across all three category tiers without exception.

Category	Return Rate	Top Return Reason
Budget	7.84%	Defective
Mid-Range	7.24%	Defective
Premium	7.02%	Defective

Table 4-9 Return Rate and Leading Return Reason by Category Tier

Q19. What is the most common return reason across all returned orders, and how is it distributed across product categories?

```
SELECT
    p.category,
    f.return_reason,
    COUNT(*) AS Total_Returns
FROM fact_orders f
JOIN dim_products p ON f.sku = p.sku
WHERE f.units_returned > 0
GROUP BY p.category, f.return_reason
ORDER BY Total_Returns DESC;
```

Finding: "Defective" is the single most common return reason overall, accounting for 9,588 returned units, ahead of "Changed Mind" (5,120), "Not As Described" (5,050), "Wrong Item" (4,011), and "Damaged In Shipping" (3,791), and this pattern holds consistently across every product category.

Return Reason	Number of Orders Returned	Share of Returns
Defective	513	36.7%
Not as Described	267	19.1%
Changed Mind	252	18.0%
Wrong Item	224	16.0%
Damaged in Shipping	204	14.6%

Table 4-4 Distribution of Orders by Return Reason

Q20. How many SKUs are currently flagged as stockout risk, and which product types are most affected?

```
SELECT
    p.product_type,
    COUNT(*) AS Stockout_SKUs
FROM subdim_inventory s
JOIN dim_products p ON s.sku = p.sku
WHERE s.stockout_risk = 1
GROUP BY p.product_type
ORDER BY Stockout_SKUs DESC;
```

Finding: A total of 831 SKUs are currently flagged at stockout risk across a total inventory volume of 230.9 thousand units, with Smartphones — the highest-volume product type — also the most affected category, compounding the revenue exposure of a potential stockout.

Product Type	SKUs at Stockout Risk
Smartphones	210
Tablets	175
Headphones	163
Laptops	158
Smartwatches	125

Table 4-5 SKUs Flagged at Stockout Risk by Product Type (Total: 831)

Q21. Which product type has the highest average defect rate, and does it also record the highest return rate?

```
SELECT
    p.product_type,
    AVG(f.defect_rates) AS Avg_Defect_Rate,
    SUM(f.units_returned) * 1.0 / SUM(f.number_of_products_sold) AS Return_Rate
FROM fact_orders f
JOIN dim_products p ON f.sku = p.sku
GROUP BY p.product_type
ORDER BY Avg_Defect_Rate DESC;
```

Finding: Smartphones record the highest average defect rate (4.82) among the five product types and simultaneously the highest return rate (7.6%), confirming a direct link between manufacturing defect rate and customer return behavior for this category.

Q22. Which product type generates the highest total number of units sold, and what is its share of total sales volume?

```
SELECT
    p.product_type,
    SUM(f.number_of_products_sold) AS Total_Units_Sold,
    SUM(f.number_of_products_sold) * 100.0 /
        (SELECT SUM(number_of_products_sold) FROM fact_orders) AS Sales_Share
FROM fact_orders f
JOIN dim_products p ON f.sku = p.sku
GROUP BY p.product_type
ORDER BY Total_Units_Sold DESC;
```

Finding: Smartphones generate the highest total sales volume, representing a 25.14% share of all units sold, ahead of Tablets (20.58%), Laptops (19.88%), Headphones (18.76%), and Smartwatches (15.64%).

4.5 Summary of Findings by Domain

Table 4-1 consolidates the headline finding from each of the twenty-two questions into a single reference table, organized by analytical domain.

Domain	Key Finding
Financial Performance	\$182.8M revenue, \$97.6M profit, 53.37% margin; Wholesale highest margin (53.9%); discount depth inversely tied to profit
Supplier Performance	Supplier 2 top profit contributor (\$21.07M); Supplier 5 weakest on reliability, quality, and revenue lost to returns (\$2.95M)
Logistics & Operations	80.4% overall on-time rate; DHL best on-time (83.45%), Blue Dart weakest (66.67%); air freight slowest and costliest mode; delivery timing has only a marginal effect on satisfaction
Returns & Inventory	"Defective" is the top return reason (9,588 units); Smartphones highest return rate (7.6%) and defect rate (4.82); 831 SKUs at stockout risk

Table 4-1 Consolidated Findings by Analytical Domain

Chapter Summary

This chapter presented the twenty-two structured analysis questions developed for the project, each with its exact T-SQL implementation and resulting finding, grouped into Financial Performance, Supplier Performance, Logistics & Operations, and Returns & Inventory. These findings — spanning revenue and margin trends, supplier reliability and defect patterns, carrier cost-speed trade-offs, and return/inventory risk — form the analytical foundation for the interactive Tableau dashboards presented in the next chapter.

5 Phase Three: Data Visualization

The third phase of the project translated the twenty-two analytical findings from Chapter Four into an interactive, stakeholder-facing visualization layer built in Tableau. This chapter introduces the visualization approach and then presents each of the five dashboards in the workbook alongside its full screenshot, walking through every chart, KPI card, and filter it contains.

5.1 Introduction

All five dashboards share a common visual language: a navy-to-teal-to-gold gradient navigation panel on the left, bearing the Digital Egypt Pioneers Initiative logo and an icon for each dashboard (Overview, Finance, Products, Suppliers, Carriers), and a consistent header bar at the top of every dashboard offering Year, Region, and Category filters. This shared structure allows a viewer to apply a single filter selection — for example, a specific year or region — and see it carried consistently across all five views, supporting the kind of cross-domain exploration that a flat spreadsheet cannot easily provide.

5.2 Supply Chain Executive Overview

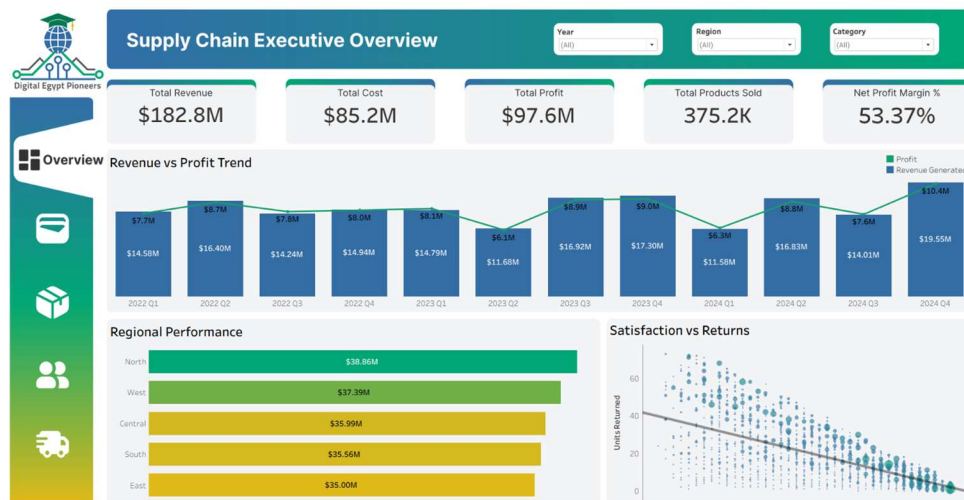


Figure 5-1 Supply Chain Executive Overview Dashboard

The Executive Overview dashboard opens the workbook with five headline KPI cards — Total Revenue (\$182.8M), Total Cost (\$85.2M), Total Profit (\$97.6M), Total Products Sold (375.2K), and Net Profit Margin % (53.37%) — giving a stakeholder an immediate read on overall business health. Below the KPI row, a Revenue vs. Profit Trend combination chart plots quarterly revenue (bars) against profit (line) from 2022 Q1 through 2024 Q4, showing a generally upward trajectory that peaks at \$19.55 million in revenue and \$10.4 million in profit during 2024 Q4.

A Regional Performance bar chart ranks the five regions by revenue, led by North at \$38.86 million, followed by West (\$37.39M), Central (\$35.99M), South (\$35.56M), and East (\$35.00M). A Satisfaction vs. Returns scatter plot in the bottom-right shows a clear downward-sloping relationship: orders with more units returned tend to cluster at lower customer-satisfaction scores. This return-driven pattern is the dominant satisfaction effect in the dataset, whereas Question 16 shows that delivery timeliness alone has only a marginal effect on satisfaction by comparison.

5.3 Finance Analysis

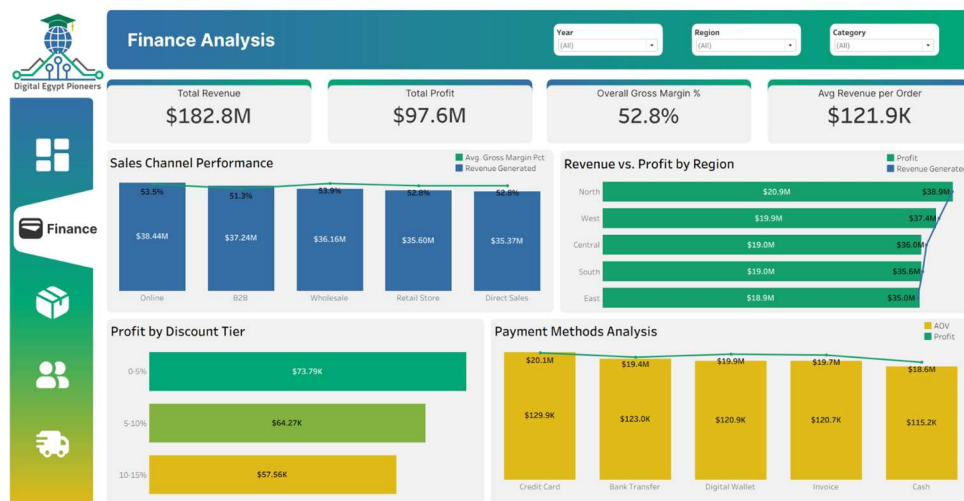


Figure 5-2 Finance Analysis Dashboard

The Finance Analysis dashboard focuses on channel- and payment-level financial detail. Four KPI cards repeat the headline Total Revenue, Total Profit, Overall Gross Margin %, and Avg Revenue per Order figures for context. The Sales Channel Performance chart combines a bar-and-line view showing Online as the top channel by revenue (\$38.44M) and among the highest in gross margin (53.5%), with Wholesale recording the single highest margin at 53.9%, followed by Retail Store (52.8%) and Direct Sales (52.8%).

A Revenue vs. Profit by Region horizontal bar chart repeats the regional breakdown with both metrics side by side for direct comparison. The Profit by Discount Tier chart makes the discount-versus-profitability trade-off explicit: profit declines steadily from \$73.79K at the 0-5% discount tier down to \$64.27K at 5-10% and \$57.56K at 10-15%. Finally, the Payment Methods Analysis chart shows Credit Card as the leading payment method by both revenue (\$20.1M) and average order value (\$129.9K), narrowly ahead of Bank Transfer, Digital Wallet, and Invoice, with Cash recording the lowest average order value (\$115.2K).

5.4 Product & Inventory Analysis

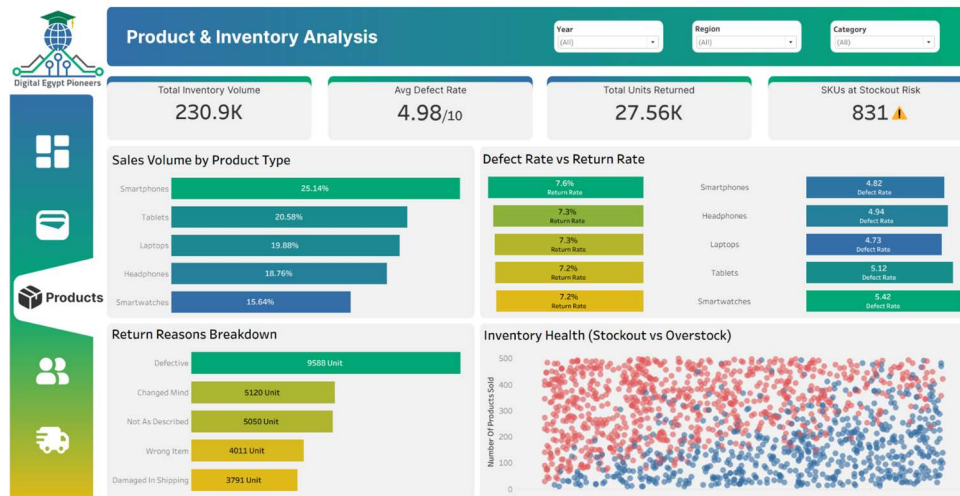


Figure 5-3 Product & Inventory Analysis Dashboard

This dashboard shifts the focus to product-level and inventory health metrics. The KPI row reports Total Inventory Volume (230.9K), Avg Defect Rate (4.98/10), Total Units Returned (27.56K), and SKUs at Stockout Risk (831, flagged with a warning indicator). The Sales Volume by Product Type bar chart shows Smartphones leading at a 25.14% share, ahead of Tablets (20.58%), Laptops (19.88%), Headphones (18.76%), and Smartwatches (15.64%).

The Defect Rate vs. Return Rate paired panel places each product type's return rate alongside its defect rate, showing Smartphones at the top on both measures (7.6% return rate, 4.82 defect rate) — visual confirmation of Question 21's finding that defect rate and return rate move together for this category. A Return Reasons Breakdown bar chart confirms "Defective" as the dominant return reason at 9,588 units, ahead of "Changed Mind" (5,120), "Not As Described" (5,050), "Wrong Item" (4,011), and "Damaged In Shipping" (3,791). An Inventory Health scatter plot contrasts products sold against stockout (red) versus overstock (blue) risk across the full product range, visually surfacing the concentration of at-risk SKUs.

5.5 Supplier Performance

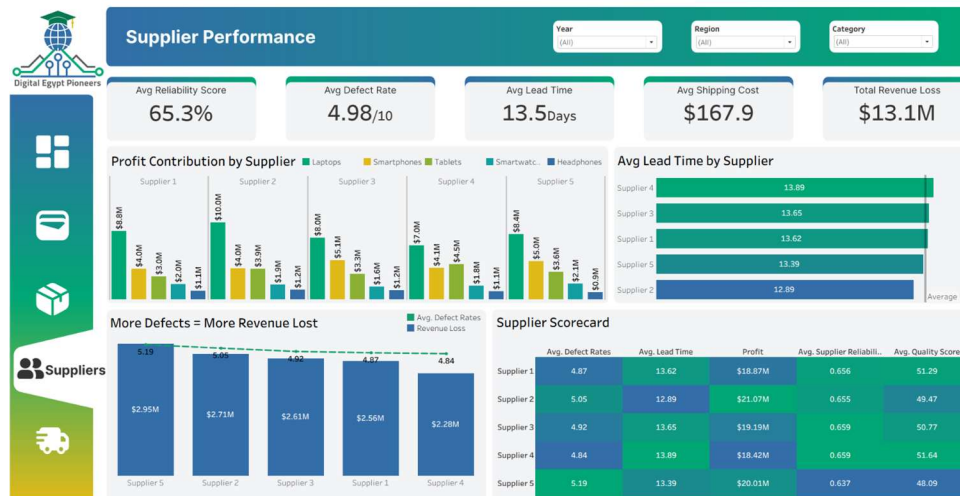


Figure 5-4 Supplier Performance Dashboard

The Supplier Performance dashboard consolidates the five-supplier comparison from Questions 11 and 12. KPI cards report an Avg Reliability Score of 65.3%, Avg Defect Rate of 4.98/10, Avg Lead Time of 13.5 days, Avg Shipping Cost of \$167.9, and Total Revenue Loss of \$13.1M. A stacked Profit Contribution by Supplier chart breaks each supplier's profit down by product category, with Supplier 2 the strongest overall contributor at \$10.0M.

An Avg Lead Time by Supplier bar chart ranks suppliers against the overall average line, showing Supplier 2 fastest at 12.89 days and Supplier 4 slowest at 13.89 days. The More Defects = More Revenue Lost combination chart directly pairs each supplier's defect rate with its associated revenue loss in descending order, from Supplier 5 (5.19 defect rate, \$2.95M lost) down to Supplier 4 (4.84 defect rate, \$2.28M lost). A Supplier Scorecard table consolidates defect rate, lead time, profit, reliability, and quality score for all five suppliers in one reference view, using green-to-yellow-to-blue heat-shaded formatting to make relative performance immediately visible at a glance.

5.6 Logistics & Carrier Performance

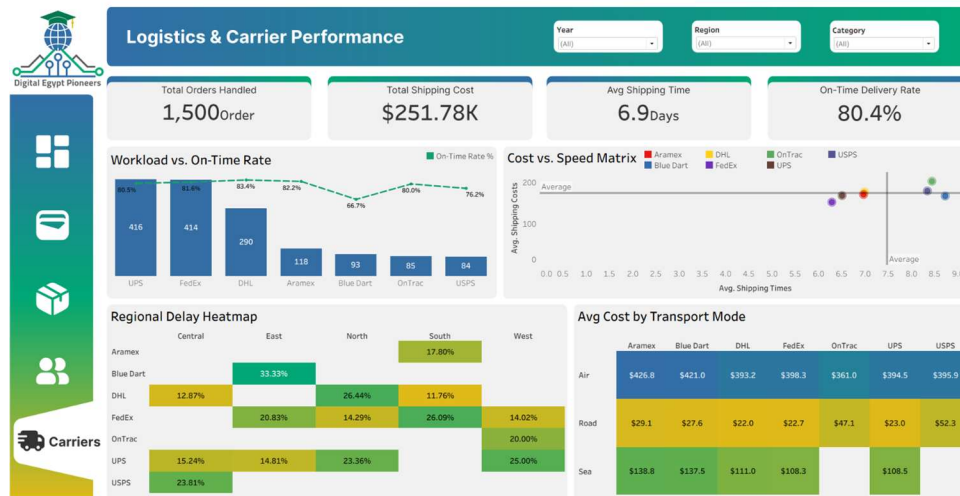


Figure 5-5 Logistics & Carrier Performance Dashboard

The final dashboard addresses carrier and shipping performance. KPI cards report Total Orders Handled (1,500), Total Shipping Cost (\$251.78K), Avg Shipping Time (6.9 days), and On-Time Delivery Rate (80.4%). The Workload vs. On-Time Rate combination chart ranks carriers by order volume (UPS 416, FedEx 414, DHL 290, Aramex 118, Blue Dart 93, OnTrac 85, USPS 84) against an on-time-rate line that dips to its lowest point at Blue Dart (66.7%) despite Blue Dart not being the lowest-volume carrier, while DHL — a mid-volume carrier — posts the highest on-time rate at 83.45%, confirming that order volume alone does not predict delivery reliability.

A Cost vs. Speed Matrix scatter plot positions all seven carriers against average-cost and average-shipping-time axes split at the overall average in both dimensions, immediately identifying which carriers sit in the favorable low-cost/low-time quadrant. A Regional Delay Heatmap cross-tabulates each carrier's delay percentage by region, exposing carrier-and-region-specific problem pairs — such as Blue Dart's 33.33% delay rate in the East — rather than a single overall carrier average. Finally, an Avg Cost by Transport Mode table breaks down cost by carrier and mode (Air, Road, Sea), confirming that air shipping costs roughly ten to fifteen times more than road or sea across every carrier in the dataset, for example \$426.8 by air versus \$29.1 by road for Aramex.

5.7 Cross-Dashboard Navigation and Filtering

Because every dashboard shares the same Year, Region, and Category filter controls and the same left-hand navigation icons, a user investigating a specific finding can move fluidly between views without losing context. For example, a stakeholder who notices Supplier 5's elevated defect rate on the Supplier Performance dashboard can filter to the corresponding region and category, then switch to the Product & Inventory Analysis dashboard to see whether the same product types show elevated stockout risk, and finally check the Logistics & Carrier Performance dashboard to see whether the same region also shows elevated carrier delays — supporting exactly the kind of cross-domain root-cause investigation that a flat spreadsheet report cannot easily provide.

Chapter Summary

This chapter presented the five-dashboard Tableau analytics suite — Executive Overview, Finance Analysis, Product & Inventory Analysis, Supplier Performance, and Logistics & Carrier Performance — with a full screenshot and detailed walkthrough of every chart and KPI in each. The consistent navigation and filter structure across all five dashboards allows a single workbook to serve financial, product, supplier, and logistics audiences alike, translating the twenty-two analysis questions of Chapter Four into an interactive, stakeholder-ready format.

6 Conclusion and Recommendations

This final chapter draws together the outcomes of the project and translates the analytical findings from the preceding chapters into a set of practical, prioritized recommendations for improving supply chain performance.

6.1 Conclusion

This project applied a complete data-analytics workflow to a 1,500-order electronics supply chain dataset, progressing from raw-data cleaning through relational data modeling, structured SQL analysis, and interactive Tableau visualization. Chapter Two established the shape and quality of the raw data; Chapter Three documented the exact cleaning steps and the star-schema model built to support efficient querying; Chapter Four answered twenty-two structured business questions with fully reproducible SQL; and Chapter Five rendered every finding into an interactive, filterable dashboard suite.

The cleaned and modeled data confirmed a financially healthy business, generating \$182.8 million in total revenue and \$97.6 million in total profit at a 53.37% gross margin, while also surfacing clear, addressable weaknesses: an average on-time delivery rate of only 80.4%, an average supplier reliability score of just 65.3%, 831 SKUs at stockout risk, and Smartphones standing out as the product category with the highest combined defect and return rate. The consistency of these findings across three independent tool chains — Excel, SQL Server, and Python — together with their clear visual representation in the five-dashboard Tableau workbook, demonstrates that the project's twenty-two analysis questions were answered with a reliable, decision-ready degree of confidence.

6.2 Recommendations

6.2.1 Short-Term Recommendations

- Prioritize corrective engagement with Supplier 5, which records the lowest reliability score (0.637), lowest quality score (48.09), and highest revenue loss from returns (\$2.95M) among the five evaluated suppliers.
- Review Blue Dart's operational performance in particular, given its comparatively low on-time delivery rate (66.67%) — the lowest among all seven carriers evaluated — and investigate the East-region delay pattern responsible for a large share of this shortfall.
- Address the 831 SKUs currently flagged at stockout risk, prioritizing Smartphones and other high-volume product types where a stockout would have the largest revenue impact.
- Investigate the root causes behind the elevated defect and return rate for Smartphones, given that "Defective" is already the single most common return reason across the entire dataset.
- Target Blue Dart's regional delay pattern in the East specifically, rather than adjusting the carrier's contract terms across all regions uniformly.

6.2.2 Long-Term Recommendations

- Establish ongoing, automated monitoring of the supplier and carrier scorecards introduced in this project, so that reliability, defect-rate, and on-time-delivery trends can be tracked continuously rather than assessed as a one-time analysis.
- Reassess discount strategy in light of the clear inverse relationship between discount depth and profit observed across all three discount tiers, ensuring that promotional discounting is targeted rather than broadly applied.
- Extend the current relational data model with additional historical data as it becomes available, allowing the fact_orders, dim_supplier, dim_carriers, and dim_products structure built in this project to scale into a permanent analytical asset rather than a one-time reporting exercise.
- Explore predictive extensions of this analysis — such as forecasting stockout risk or return likelihood — as a natural next phase building on the descriptive and diagnostic foundation established here.
- Standardize carrier-name and categorical-text capture at the point of data entry in source systems, reducing the amount of manual standardization required in future cleaning cycles.

Chapter Summary

This chapter concluded the documentation by summarizing the project's outcomes against its original objectives and translating the analytical findings into concrete short-term and long-term recommendations. Together with the preceding chapters, it completes the record of a full analytics lifecycle — from raw data to relational model to structured analysis to interactive visualization — applied to the electronics supply chain dataset.